IDX G9 COMPUTER SCIENCE S STUDY GUIDE

ISSUE S2 FINALS

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This comprehensive review guide covers essential topics in Information Technology, including Database Fundamentals, Computer Troubleshooting, Information Security, Privacy and Ethics, Artificial Intelligence, and System Software. Use this guide to prepare for your final exam by focusing on key concepts, practical examples, and exam tips.

Section 1: Database Fundamentals

1.1 Database Concepts

A **database** is an organized collection of data, typically stored and accessed electronically from a computer system. Databases are managed by a **Database Management System (DBMS)**, which provides tools for data creation, manipulation, and retrieval.

Data Hierarchy

- **Character**: The smallest unit of data (e.g., 'A', '7', '@').
- **Field (Attribute)**: A single data category, represented as a column in a table (e.g., "Name", "Age").
- **Record (Tuple)**: A single row in a table, representing a complete data entry (e.g., [101, "Alice", 25]).
- **Table (Relation)**: A collection of related records, organized in rows and columns (e.g., "Employee" table).
- **Database**: A collection of interrelated tables (e.g., "Company_Database").

DBMS Models

Model	Structure	Relationship Type	Use Case	Example
Hierarchical	Tree-like (parent-child)	One-to-many	Organizational charts	IBM IMS
Network	Graph-like	Many-to-many	Complex relationships	CODASYL DB
Relational	Tables (rows & columns)	Foreign keys	Most modern applications	MySQL, PostgreSQL, Oracle

- **Hierarchical Model**: Data is organized in a tree structure with a single root. Each child has one parent, limiting flexibility but efficient for hierarchical data.
- **Network Model**: Allows multiple parent-child relationships, suitable for complex data but harder to maintain.
- **Relational Model**: Most widely used, organizes data into tables linked by foreign keys. Supports SQL for querying.

1.2 SQL Basics

SQL (Structured Query Language) is used to interact with relational databases. It supports creating, reading, updating, and deleting data (CRUD operations).

Creating Tables

```
CREATE TABLE Employee (
    EmployeeID INT PRIMARY KEY,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50),
    Salary FLOAT,
    HireDate DATE,
    DepartmentID INT,
    FOREIGN KEY (DepartmentID) REFERENCES Department(DepartmentID)
);
```

- **PRIMARY KEY**: Uniquely identifies each record.
- **FOREIGN KEY**: Links tables by referencing another table’s primary key.
- **NOT NULL**: Ensures a column cannot have null values.

Common Data Types

- **INT**: Whole numbers (e.g., 100, -5).
- **FLOAT**: Decimal numbers (e.g., 123.45).
- **VARCHAR(n)**: Variable-length text with a maximum length of n (e.g., VARCHAR(50) for names).
- **CHAR(n)**: Fixed-length text (e.g., CHAR(1) for gender).
- **DATE**: Stores dates (e.g., '2025-06-14').
- **TEXT**: Large text data (e.g., descriptions).

SQL CRUD Operations

Operation	Syntax	Example
SELECT	`SELECT columns FROM table WHERE condition;`	`SELECT FirstName, Salary FROM Employee WHERE DepartmentID = 3;`

Operation	Syntax	Example
INSERT	`INSERT INTO table (columns) VALUES (values);`	`INSERT INTO Employee (EmployeeID, FirstName, Salary) VALUES (101, 'Alice', 60000);`
UPDATE	`UPDATE table SET column = value WHERE condition;`	`UPDATE Employee SET Salary = 65000 WHERE EmployeeID = 101;`
DELETE	`DELETE FROM table WHERE condition;`	`DELETE FROM Employee WHERE Salary < 30000;`

- **SELECT**: Retrieves data. Use `*` to select all columns.
- **INSERT**: Adds new records. Specify columns and values.
- **UPDATE**: Modifies existing records based on a condition.
- **DELETE**: Removes records based on a condition.

Advanced SQL Queries

- **ORDER BY**: Sorts results.
- `SELECT FirstName, Salary FROM Employee ORDER BY Salary DESC;`
- **LIMIT**: Restricts the number of rows returned.
- `SELECT * FROM Orders LIMIT 5;`
- **Aggregate Functions**:
  - `COUNT(*)`: Counts total rows.
  - `MAX(column)`: Finds the maximum value.
  - `MIN(column)`: Finds the minimum value.
  - `AVG(column)`: Calculates the average.
  - `SUM(column)`: Calculates the sum.
- `SELECT DepartmentID, AVG(Salary) AS AvgSalary FROM Employee GROUP BY DepartmentID;`
- **JOIN**: Combines data from multiple tables.
- `SELECT Employee.FirstName, Department.DepartmentName`
- `FROM Employee`
- `INNER JOIN Department ON Employee.DepartmentID = Department.DepartmentID;`
  - **INNER JOIN**: Returns matching records from both tables.
  - **LEFT JOIN**: Returns all records from the left table and matching records from the right.
  - **RIGHT JOIN**: Returns all records from the right table and matching records from the left.
- **ALTER TABLE**: Modifies table structure.
- `ALTER TABLE Employee ADD COLUMN Email VARCHAR(100);`

Section 2: Computer Troubleshooting

2.1 BIOS Basics

The **BIOS (Basic Input/Output System)** is firmware that initializes hardware during the boot process and provides runtime services for operating systems.

- **Functions**:
  - Performs **POST (Power-On Self-Test)** to check hardware functionality.
  - Loads the bootloader to start the operating system.

    - Manages basic hardware settings (e.g., date/time, boot order).
- **Accessing BIOS**:
    - Press keys like **F2**, **F10**, **F12**, or **DEL** during startup (varies by manufacturer).
    - Common settings:
        - **Power Settings**: Control power management (e.g., sleep modes).
        - **Date/Time**: System clock settings.
        - **Boot Order**: Prioritizes boot devices (e.g., SSD, USB, CD).

2.2 Boot Process

1. **BIOS Activation**: BIOS initializes and checks hardware.
2. **POST**: Tests critical components (CPU, RAM, GPU).
3. **Bootloader**: Loads the operating system into RAM.
4. **OS Initialization**: Loads drivers and system configurations.

2.3 Troubleshooting Boot Issues

Issue	Symptoms	Solution
No Boot	Blank screen, no power	Check power supply, reseat RAM/GPU, test with minimal hardware.
Blue Screen (BSOD)	System crashes with error code	Boot in **Safe Mode (F8)**, update drivers, run diagnostics (e.g., chkdsk).
Corrupt OS	Fails to load OS	Use recovery USB/CD, repair OS, or reinstall.
Slow Boot	Long loading times	Disable unnecessary startup programs, run virus scans, defragment disk.
Boot Loop	Restarts repeatedly	Recover **Master Boot Record (MBR)**, check for malware, repair OS.

- **Safe Mode**: Loads minimal drivers for troubleshooting.
- **Recovery Media**: Boot from USB/CD to repair or reinstall the OS.
- **Virus Scans**: Use tools like Windows Defender or third-party antivirus.

Section 3: Information Security

3.1 Encryption Basics

Encryption converts plaintext into ciphertext to protect data. **Decryption** reverses the process.

Types of Encryption

Type	Key Usage	Algorithms	Use Case
Symmetric	Same key for encryption/decryption	AES, DES, 3DES	Fast encryption (e.g., file storage)
Asymmetric	Public key (encrypt), private key (decrypt)	RSA, ECC	Secure key exchange, digital signatures

- **Symmetric Encryption**:
    - **DES (Data Encryption Standard)**: 56-bit key, outdated.
    - **AES (Advanced Encryption Standard)**: 128/192/256-bit keys, widely used.
    - Example: Encrypting a file with AES-256.
- **Asymmetric Encryption**:
    - Slower but secure for key exchange.
    - Example: RSA for secure email communication.
- **Caesar Cipher** (Historical Example):
    - Shifts letters by a fixed number (e.g., Shift = 3: A → D, B → E).
    - Decryption: Reverse the shift.
    - Weak due to limited key space (26 possible shifts).

3.2 Security Protocols

- **TLS/SSL (Transport Layer Security/Secure Sockets Layer)**:
    - Secures communication over networks (e.g., HTTPS websites).
    - Uses asymmetric encryption for key exchange and symmetric for data transfer.
- **VPN (Virtual Private Network)**:
    - Creates an encrypted tunnel for secure remote access.
    - **Types**:
        - **Voluntary Tunneling**: Client initiates the connection.
        - **Compulsory Tunneling**: Network enforces the connection.
    - **Encryption Methods**: Combines symmetric (data transfer) and asymmetric (key exchange).

Section 4: Information Privacy and Ethics

4.1 Privacy Concerns

- **Personal Information Exposure**: Risks include leaking photos, addresses, or IDs.
- **Web Tracking**: Websites track browsing history via cookies or scripts.
- **Cookies**:
    - Store user data (e.g., login sessions, preferences).
    - Types: Session cookies (temporary), persistent cookies (long-term tracking).

4.2 Privacy Protection

- **Clear Browser History/Cookies**: Regularly delete to reduce tracking.
- **Avoid Untrusted ActiveX Controls**: Can execute malicious code.
- **Use VPNs**: Encrypts traffic to protect against snooping.
- **Strong Passwords**: Use complex, unique passwords with two-factor authentication (2FA).

4.3 Computer Ethics

- **Five "NOTs"**:
    1. Don’t harm others with malicious code (e.g., viruses).
    2. Don’t interfere with others’ work (e.g., hacking systems).
    3. Don’t steal data or resources.

    4. Don’t use unpaid proprietary software (e.g., pirated software).
    5. Don’t appropriate others’ intellectual output (e.g., plagiarism).
- **One "ALWAYS"**:
    - Use computers with consideration and respect for others.

Section 5: Artificial Intelligence

5.1 AI Fundamentals

- **Weak AI (Narrow AI)**: Designed for specific tasks.
    - Examples: Siri, Alexa, recommendation systems.
- **Strong AI (General AI)**: Theoretical human-level intelligence capable of any intellectual task.
    - Not yet achieved; remains a research goal.

Key Elements of AI

- **Processing Power**: High-performance hardware (e.g., GPUs, TPUs).
- **Quality Data**: Large, clean datasets for training.
- **Effective Algorithms**: Machine learning models, neural networks.

5.2 Machine Learning Types

Type	Learning Method	Example Application
Supervised	Uses labeled data (input-output pairs)	Spam detection, image classification
Unsupervised	Finds patterns in unlabeled data	Customer segmentation, anomaly detection
Reinforcement	Learns via rewards/penalties	Game AI, robotics

- **Supervised Learning**:
    - **Classification**: Predicts discrete categories (e.g., spam vs. not spam).
    - **Regression**: Predicts continuous variables (e.g., house prices).
- **Unsupervised Learning**: Clusters data or reduces dimensionality (e.g., grouping customers by behavior).
- **Reinforcement Learning**: Agent learns by interacting with an environment (e.g., AlphaGo learning through games).

5.3 Neural Networks

- **Structure**:
    - **Input Layer**: Receives raw data.
    - **Hidden Layers**: Process data through transformations.
    - **Output Layer**: Produces final predictions.
- **Deep Learning**: Neural networks with many layers (deep, neural networks).
    - Example: Convolutional Neural Networks (CNNs) for image recognition.
- **Training**:
    - **Epochs weights/biases** using backpropagation.
    - More epochs improve accuracy but risk overfitting.

5.4 AI Milestones

Year	Event	Significance
1950	Turing Test proposed	Framework for evaluating AI intelligence
1997	Deep Blue defeats chess champion	AI excels in rule-based games
2016	AlphaGo defeats Go champion	AI masters complex, intuitive strategy
2023	Rise of LLMs (e.g., ChatGPT)	Natural language processing reaches new heights

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Section 6: System Software

6.1 Types of System Software

- **BIOS**: Initializes hardware and starts the OS.
- **Operating System**: Manages hardware/software resources (e.g., Windows, Linux, macOS).
- **Utilities**: Tools for system maintenance (e.g., antivirus, disk cleanup).
- **Device Drivers**: Enable communication between OS and hardware (e.g., printer drivers).
- **Language Translators**: Convert code to machine language (e.g., compilers, interpreters).

6.2 Five Generations of Programming Languages

Generation	Type	Example	Characteristics
1st (1GL)	Machine Language	Binary (0s and 1s)	Hardware-specific, error-prone
2nd (2GL)	Assembly Language	MOV, ADD	Symbolic, still hardware-specific
3rd (3GL)	High-Level Language	C, Java, Python	Human-readable, portable
4th (4GL)	Very High-Level Language	SQL, MATLAB	Domain-specific, closer to human language
5th (5GL)	Natural Language	AI-based query systems	Conversational, experimental

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Final Exam Tips

- **Database Fundamentals**:
  - Practice writing **SQL queries** (SELECT, JOIN, GROUP BY, ORDER BY).
  - Understand table relationships and normalization.
- **Computer Troubleshooting**:
  - Memorize the **boot process** and common boot issue fixes.
  - Know how to access and configure **BIOS**.
- **Information Security**:
  - Differentiate **symmetric vs. asymmetric encryption**.
  - Understand **TLS/SSL** and **VPN** use cases.
- **Privacy and Ethics**:
  - Review privacy risks (e.g., cookies, data leaks).
  - Memorize the **five NOTs** and **one ALWAYS** of computer ethics.
- **Artificial Intelligence**:
  - Understand **AI types** (weak vs. strong) and **machine learning types**.

    - Know key **neural network** concepts (layers, training).
- **System Software**:
    - Differentiate types of **system software**.
    - Understand the **five generations** of programming languages.

Good Luck on Your Exam!

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